

AES DESIGN REVIEW CRITERIA

Peer Review / Design Calculator

An AES wastewater system design must be reviewed and approved by Environment Technology (ET) and the AES Design Calculator digitally signed. This signed AES Design Calculator is used by councils when processing consent applications as evidence that the design has been peer reviewed by ET, and then used by ET when supplying the correct AES components ordered by an installer.

Overview of Design Information Required

An AES wastewater system design needs to include detailed construction information for installers, the necessary information for ET to review the design and detail for Council compliance. This is a summary of the information required, explained in more detail below:

- AES Calculator
- Septic tank information
- Site plan
- Plan of AES bed (editable technical drawings are available on www.et.nz/resources/technical-drawings)
- Cross section of the AES bed (editable technical drawings are available on www.et.nz/resources/technical-drawings)
- AES bed venting

Design Information for Installers

Any approved design must include detailed construction drawings with all the information on the plan(s) and cross sections necessary for an AES accredited but not necessarily experienced drainlayer to be able to construct the AES system exactly as the designer intended.

ET does not peer review the site and soil assessment, and the digital signature is applied to a file the Design Calculator and plans only. It is therefore helpful if the plans are separated from the site report, or are in one part of the report only. It is also useful if key site specific information is mentioned either in note or diagram form on the installation plans as well as within the site report, as frequently it is the plans only that an installer refers to. The inclusion of the AES Design Calculator provides detail on how the system is sized and the estimated sand quantity necessary for construction.

This construction file can be attached to the site and soil assessment and any other information required by the council for council consent processing.

Key information that needs to be addressed in an AES design in more detail:

1. AES Calculator

- For the Design Loading Rate (DLR) use the secondary treated effluent DLR rate per AS/NZS 1547, Auckland Council's TP58 or GD06, or other council guideline, or a lower DLR based on the designer's knowledge of specific soils. Note that as the OSET- NTP AES test results were an order of magnitude better than that for Secondary Treatment quality.
- If the AES system proposed collects the effluent from a lined AES bed for dispersal via pump, the AES bed size is based on the minimum sand coverage requirements of the AES pipes, in a lined bed AES calculator and a Design Irrigation Rate (DIR) used for dispersal appropriate to the soil type at the site.

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- Installer details can be left blank if not known at time of design.

In the document for review the following needs to be shown or stated:

2. Septic tank

Location of septic tank in relation to house, boundaries if close by and AES bed. Provide level information if the AES bed is constrained by elevated ground water. Outlet filters are not required but if they are desired by either the designer or the council, attention must be paid to the consequences of them becoming blocked if they are not maintained. This is only an issue when venting of the AES bed is to exhaust at the terminal vent on the building. We do not recommend this method of venting as the possibility exists of smell emanating from this vent. If you wish to install an outlet filter in these circumstances there must be an alternative air path around the septic tank which is a better option whether a filter is fitted or not.

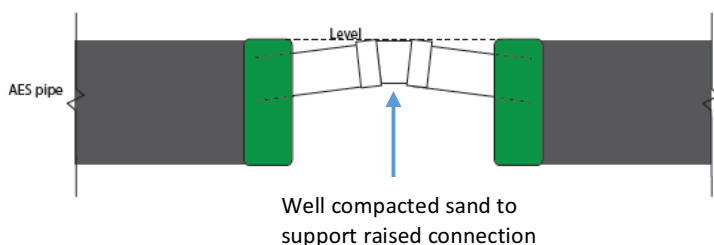
3. Site plan

- The dimensioned position of septic tank relative to building and AES bed.
- Position of the AES bed.
- Position of high and low vents with diameters noted on the plan.
- Relative levels of the pipework connecting the various components, especially if ground water or relatively impermeable layers within the proposed disposal bed area are identified.

4. Plan of AES bed

- The number of rows. Note the maximum length of a single AES row is 30 metres.
- The number of pipes per row.
- The separation between rows of 150mm minimum and 300mm from the outside rows to the walls of the excavation.
- An AES bed extension if that is required.
- A note that AES specification sand is required.
- The planned serial nature of the inter-connection of the AES pipe rows. A minimum of two rows is preferred, as this enhances the 'tidal' effect in the first row and promotes good biofilm growth. The plan below shows a three row serial slayout. If a bed is designed for installation to follow the contour on a slope or if other site constraints suit a long, thin bed area the two rows can be connected inline with a straight raised connector.

Example of a straight raised connector:



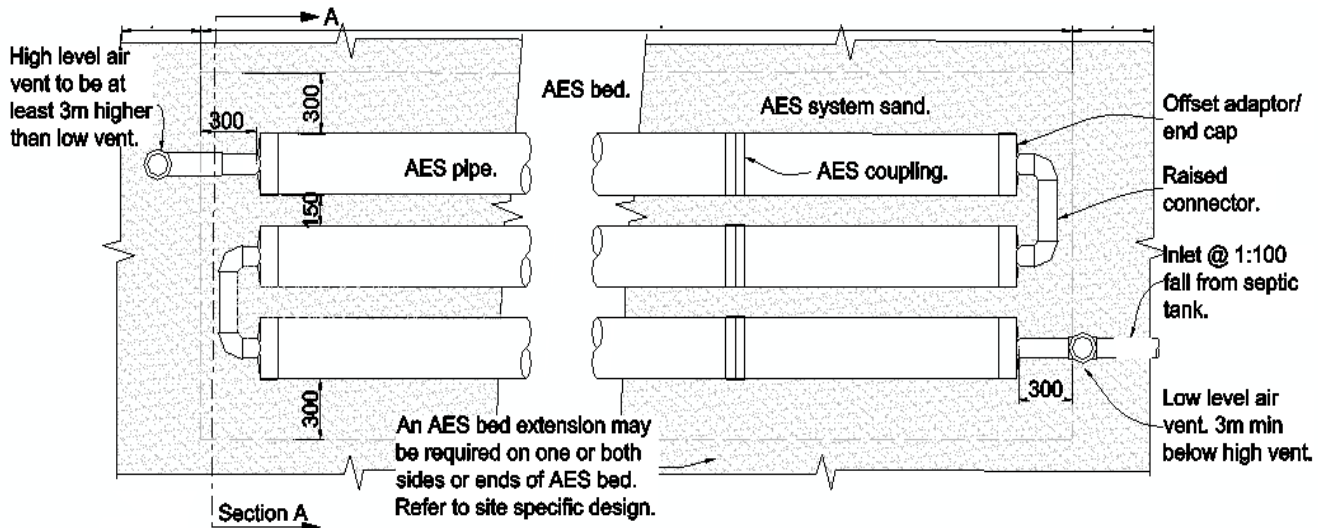
- Topsoil coverage showing a minimum of 150mm to existing surface levels. Topsoil from the site is generally appropriate. There is no maximum topsoil coverage – this can be dictated by the practicality of installation.
- Crowning of the topsoil cover to shed rain water beyond the edge of the bed, and any cutoff drains to divert other flowing surface or subsurface water away from the bed.
- The intended surface finish. Grass cover to the bed is ideal as the close root structure will assist to shed

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rain water.

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Example of a plan view:

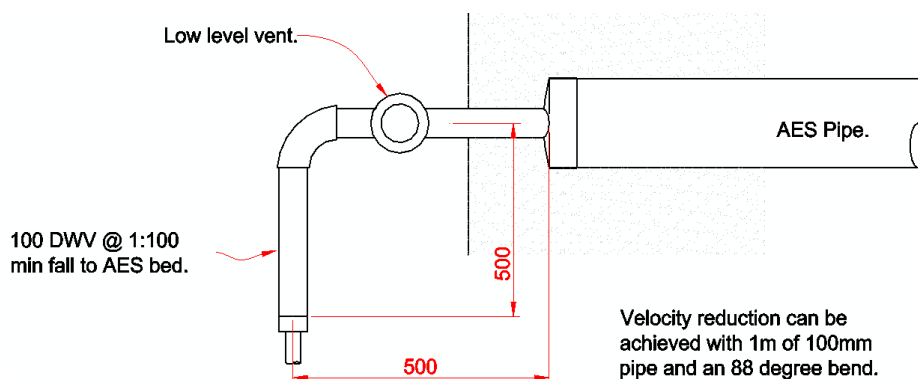


Editable templates are available on our website <http://www.et.nz/resources/technical-drawings/>

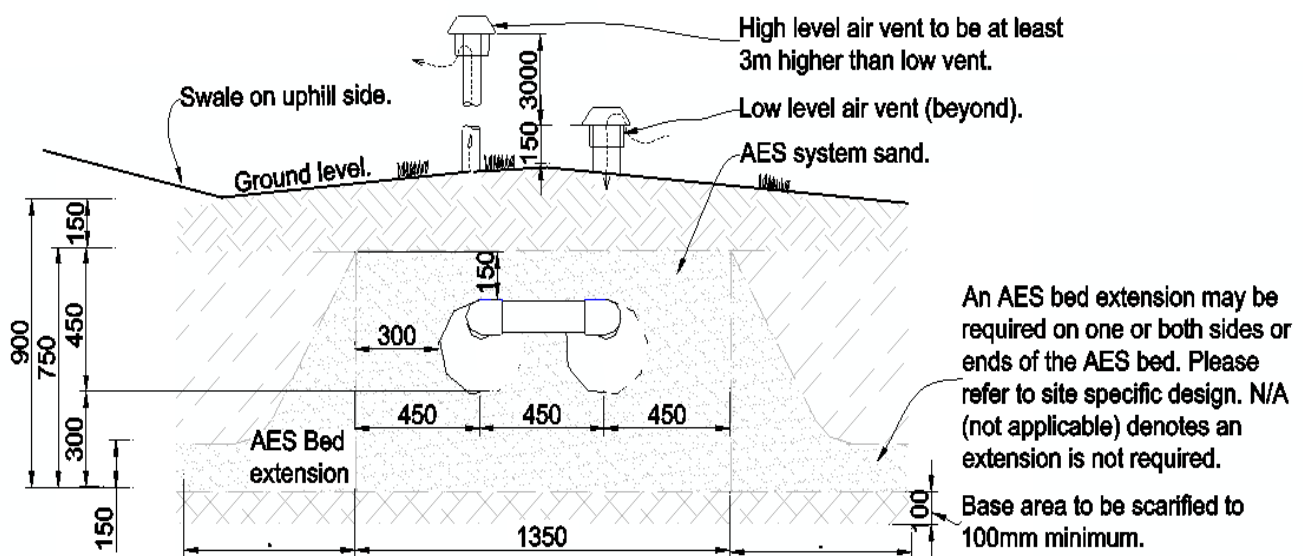
5. Cross Section of AES bed

- For Category 3-4 soils – loams and clay loams - the base of the bed needs to be scarified to 100mm with a toothed bucket or small ripper.
- For category 5-6 soils – light clay, medium clay and heavy clay soils - the base of the bed needs to be further scarified/ ripped to a depth of 200mm.
- If a sand bed extension is required include the dimensions of this on both the plan and cross-section views.
- If there is the potential for flowing surface water this must be diverted from the land application area either by a swale, cut-off trench and/or mounding of the bed top surface.
- Ensure sand coverage minimums are included, either in section or plan views or separately.
- If effluent is pumped into the AES bed, a velocity reducer must be installed to reduce turbulence inside the AES pipes – include a diagram of this reducer.

Example of a velocity reducer:



Example of a cross section:



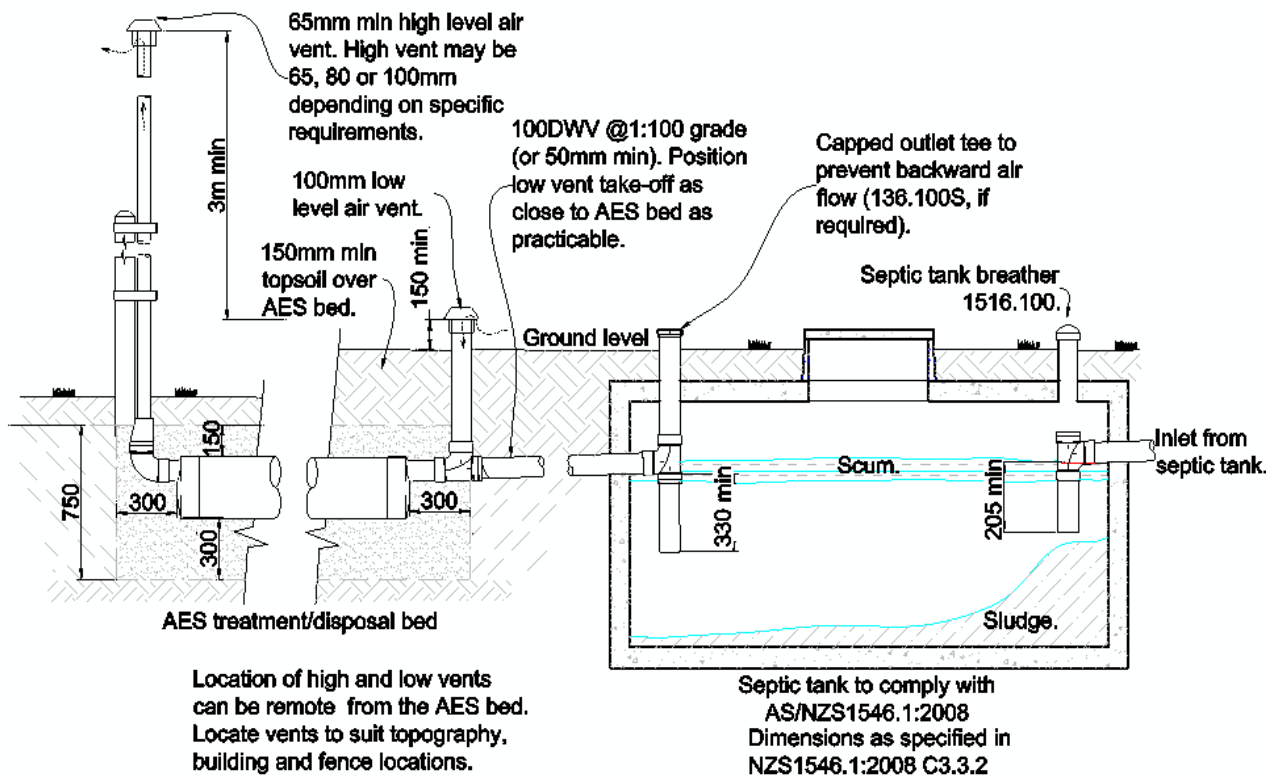
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6. Venting of the AES bed

- You must ensure the AES bed has a high air outlet vent and a low air entry vent shown in the design.
- The low vent should preferably be connected at the inflow end of AES system and be dimensioned to be a minimum of 150mm above ground in a protected position away from mowers or vehicles and above snowfall level if that is an issue.
- The high vent must be a minimum of three metres vertically above the level of the low vent and be connected preferably at the terminal end of the AES pipe system. See also septic tank notes above. If there are an even number of rows both high and low vents will be at the same end of the bed as the effluent inlet pipe, and the venting pipework can be installed in the same trench as the influent line from the septic tank, if appropriate.
- 100mm DWV pipework is appropriate underground. High level air exhaust venting above ground can be a minimum of 2 x 50mm diameter vents if installed in internal framing or 1 x 65mm minimum diameter external vent. 80mm and 100mm vents may also be used.
- Vents with preferably their own separate pipework can be located any distance from their connection to the AES system. However all connecting venting pipework must fall back to the AES bed and be noted on the plan in this manner so that no water trap that could block air flow through the system can form through condensation.
- All vents are to be kept free of vegetation.

Please refer to Technical Paper TP#1: AES System Venting for further information.

Example of a venting drawing:



7. General

- The slope of the site needs to be shown on the plan by either contours or indicated with arrow(s)
- It is helpful to see diagrams of the installation of raised connectors joining the rows similar to the following diagrams.

Example of Raised Connector:

